



kokchun giang

conceptual business-
centric modeling to more
technical **logical**
modeling

the **data modeling** journey for transactional data

business requirements

stakeholder interviews,
identify key business
processes

entities & relationships

define main objects
(entities) in the system and
how they relate to each
other

conceptual model

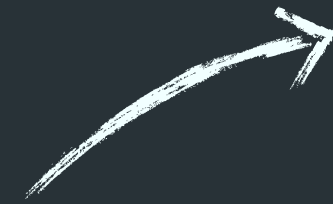
create high-level entity-
relationship diagram
(ERD), cardinality is
defined

physical model

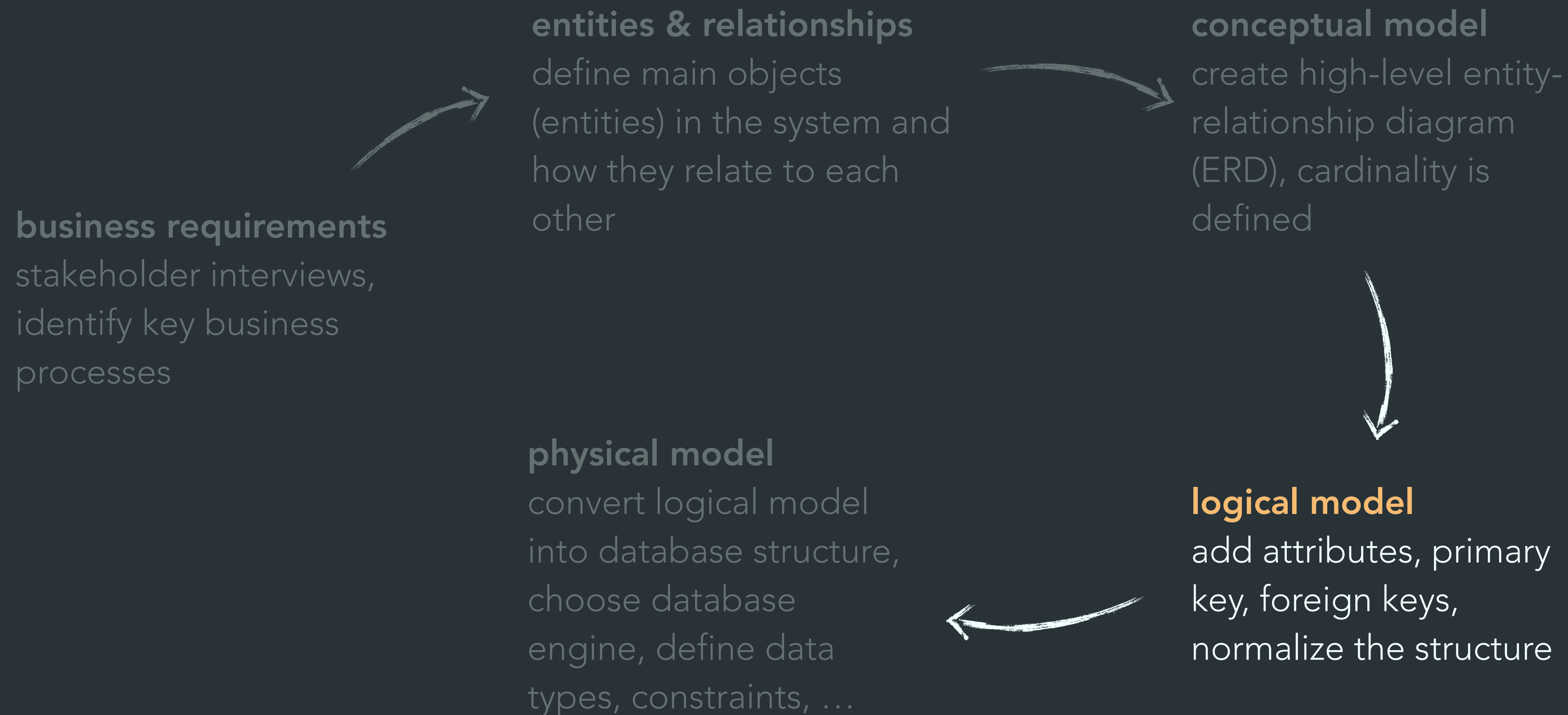
convert logical model
into database structure,
choose database
engine, define data
types, constraints, ...

logical model

add attributes, primary
key, foreign keys,
normalize the structure



the **data modeling** journey for transactional data



remember the **business requirements** for ezecream

customers should be able to browse and order ice cream flavors online

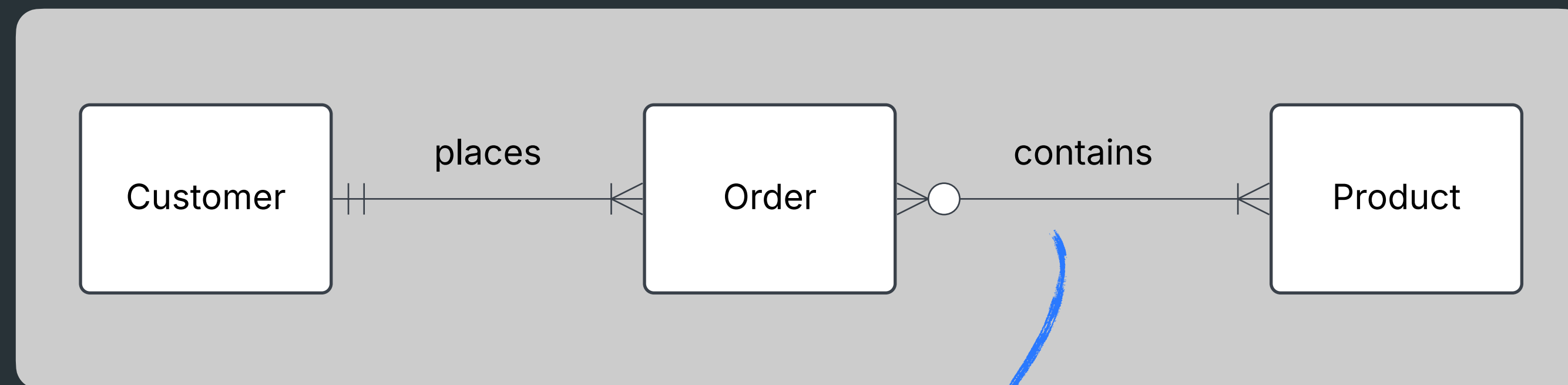
each order should contain one or more ice cream flavors

the system should store order details, including order date and total price

customers should provide their name, contact details, and delivery address

each ice cream flavor should have a name, price, and availability status

a **conceptual ERD** for ezecream using crows foot notation



many-to-many relationship,
can't be implemented
directly

introduce a
composite entity
OrderItem in between a
many-to-many

replaces a many-to-
many relationship
with two one-to-
many relationships

it acts as a bridge
table with foreign
keys that references
primary keys in the
related tables

identify the **entities & relationships** from the requirements

customers should be able to browse and order ice cream flavors online

each order should contain one or more ice cream flavors

the system should store order details, including order date and total price

attributes in Customer entity

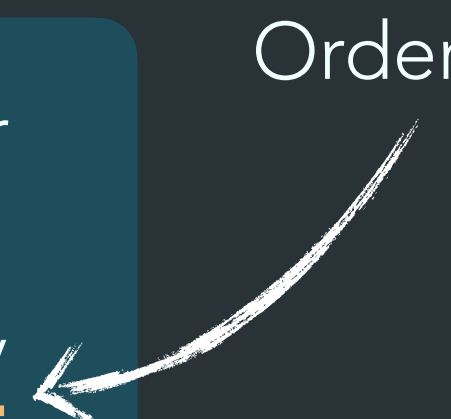
customers should provide their name, contact details, and delivery address

attributes in Product entity

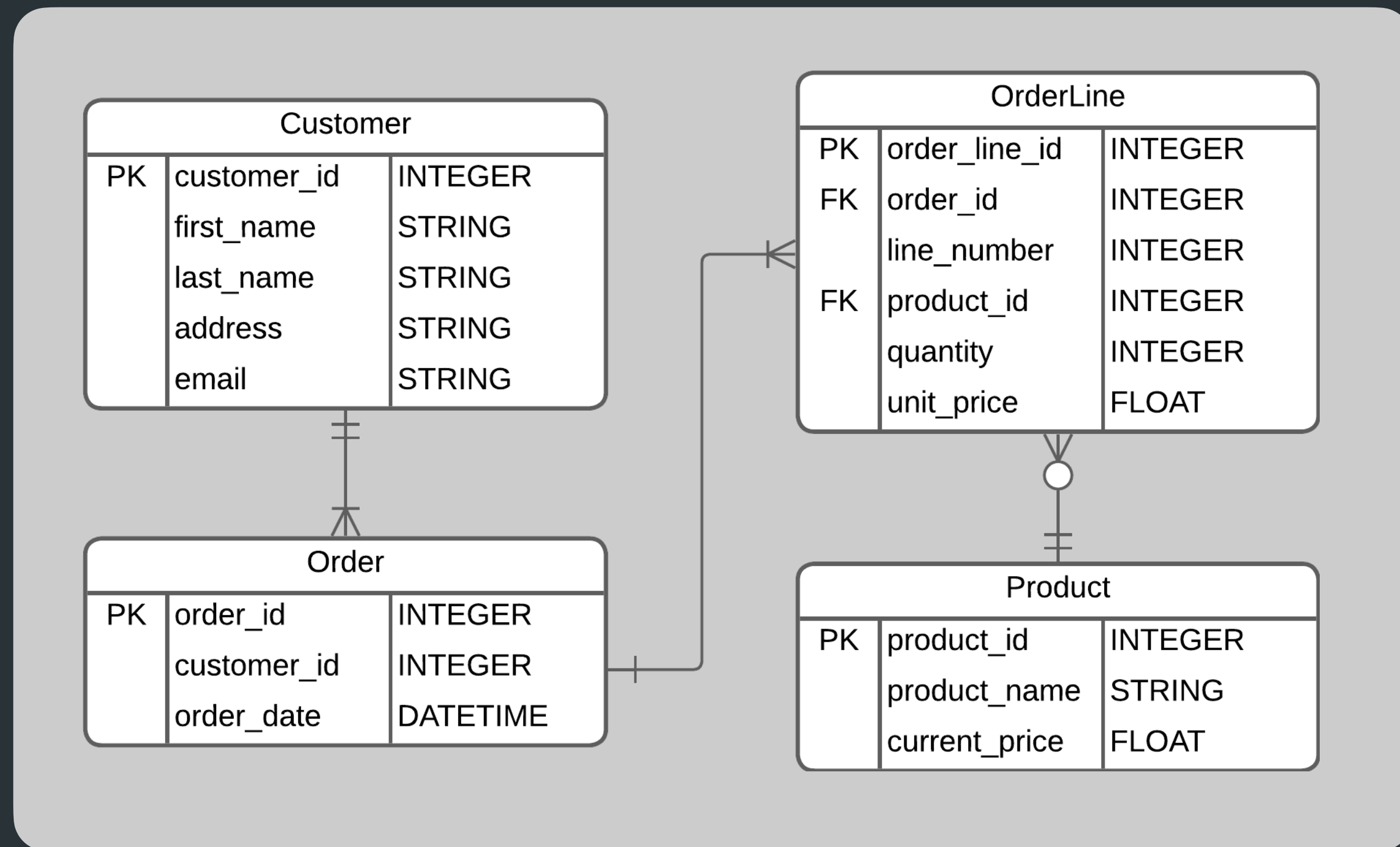
each ice cream flavor should have a name, price, and availability status

attributes in Order entity

OrderLine



the **logical data model** for ezecream



difference from conceptual model is that logical model also contains **attributes, keys & preliminary data types**

break up many-to-many to composite entity with two one-to-many

removed relationship labels

this makes the model more **specialized** while conceptual model reflects the real world more generally

the **relation schema** for ezeccream

in logical ERD phase, another way to
represent entities and its attributes is a
relation schema

attributes

Customer(customer_id (PK), first_name, last_name, address, email)

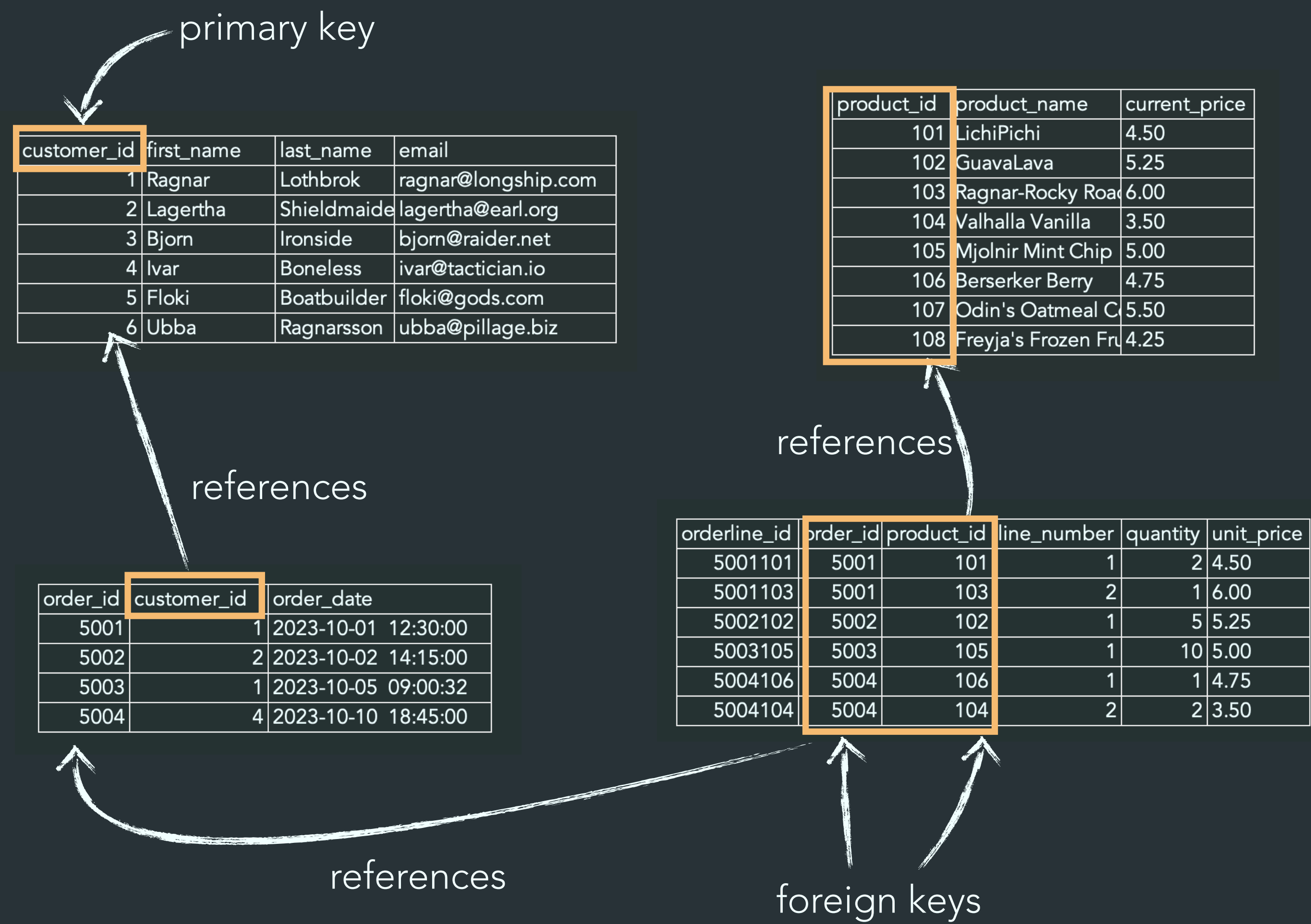
Order(order_id (PK), customer_id (FK), order_date)

keys

OrderLine(order_line_id (PK), order_id (FK), product_id (FK), line_number, quantity, unit_price)

Product(product_id (PK), product_name, current_price)

the **actual tables** could look like this in the end



foreign keys in the bridge table references primary keys in the related tables

can you figure out how much Ragnar Lodbrok needs to pay in total?

major **components** in conceptual vs logical ERD

	Conceptual	Logical
Relationship Labels	Y	
Entity Names	Y	Y
High-level Relationships	Y	Y
Cardinality of Relationships	Y	Y
Optionality of Relationships	Y	Y
Attributes		Y
Attribute Preliminary Data		Y
Primary keys		Y
Foreign keys		Y